

sell sell marketplace updates - 11042026

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What not done.



What part of the marketplace is still not done

1. Chat teaser UX is not finished

This is the main visible gap.

What's working:

- ambiguity detection works
- teaser items are returned
- clarification flow works

What's not done:

- proper horizontal teaser card layout in chat-interface.tsx
- quick-reply chips like budget / brand suggestions
- cleaner mobile rendering for the ambiguity flow

So the logic is there, but the **front-end experience is still unfinished**.

2. Local Docker/browser startup flow is not fully finalized

Your core app works, but the Dockerized local runtime still needed final confirmation around:

- app startup after migrations
- browser accessibility at localhost:3000
- entrypoint behavior

So this is not a product feature gap, but it is still an **environment/runtime gap**.

3. Full production verification is not fully closed



You reached local readiness, but the last mile still needs:

- final healthy local browser verification
- production deployment
- production smoke test after deploy

So marketplace code is largely there, but **end-to-end production validation is not completely closed yet.**

4. Broader category intelligence is still limited

Cars are clearly the most advanced flow.

What's done well:

- vehicle schema
- vehicle extraction
- vehicle-aware search
- ontology around cars
- session-based refinement

What's still not fully done:

- similar structured schemas for phones, property, furniture, etc.
- category-specific extraction logic beyond vehicles
- category-specific ranking logic beyond the stronger vehicle path

So the marketplace is not yet equally intelligent across all categories.

5. Seller-side operational features may still be basic

From what we focused on, the AI marketplace core improved a lot, but these areas were not deeply completed in this workstream:



- seller quality scoring
- fraud/risk checks
- pricing recommendations for sellers
- listing quality scoring before publish
- trust / verification layer for buyers

That means the **transaction trust layer** is still thin.

6. Analytics / product observability is not complete

You now have strong behavior, but not enough measurement around it.

What seems missing:

- search success tracking
- clarification conversion rate
- zero-result monitoring
- abandonment monitoring
- query-to-click / query-to-contact funnel

So the marketplace can behave smartly, but you still do not fully measure whether it is actually winning.

What AI monitoring is not implemented yet

This is the big one. Since the system uses AI, you need a monitoring layer beyond normal app logs.

1. Prompt / response monitoring

You likely do not yet have a proper audit trail for:

- prompt sent to Ollama
- raw model response
- parsed JSON result



- parsing failures
- fallback usage

Missing monitoring:

- how often extraction fails
- how often the model returns malformed JSON
- which prompts are unstable
- which categories are most error-prone

2. AI accuracy monitoring

You are using AI for understanding and extraction, but you do not yet appear to have metrics for:

- intent extraction accuracy
- structured field extraction accuracy
- ontology hit usefulness
- ranking relevance quality

Missing monitoring:

- was the extracted make/model/year correct
- did the AI choose the right category
- did clarification improve the result set
- did the user click something relevant after the AI answer

3. Fallback monitoring

Your system has fallbacks, which is good, but you need to measure them.

Missing monitoring:

- how often regex fallback is used instead of LLM output
- how often legacy text fallback is used instead of structured filtering
- how often ambiguity flow triggers
- how often Redis memory is unavailable and bypassed



Without this, you cannot tell whether the AI path is truly working or you are silently surviving through fallback logic.

4. Search quality monitoring

For an AI marketplace, this is critical.

Missing monitoring:

- query → results count
- query → click-through
- query → no click
- query reformulation rate
- query needing clarification
- semantic result quality vs structured result quality

This tells you whether the AI is actually helping users find things.

5. Model performance monitoring

You also need infrastructure-level AI monitoring.

Missing monitoring:

- Ollama response latency
- timeout rate
- token/output size
- model failure rate
- concurrency slowdowns
- queue buildup under load

This matters because AI systems often look fine at low load and degrade badly in real usage.

6. Drift monitoring



You are using AI on marketplace data, which changes over time.

Missing monitoring:

- new listing patterns the extractor does not understand
- new brands/models not covered by ontology
- shifts in language usage
- category drift between training assumptions and live data

Example:

if users start searching in new Arabic slang or mixed Arabic-English phrases, your AI may degrade silently.

7. Human review loop

You do not yet seem to have a review mechanism for AI mistakes.

Missing monitoring/process:

- sample bad AI outputs
- review incorrect extraction cases
- review poor ranking cases
- create correction datasets from real failures

Without this, the system cannot improve systematically.

8. Safety / hallucination monitoring

Even if your AI is mostly bounded, you still need to watch for:

- invented attributes
- wrong brand/model extraction
- misleading recommendations



- overconfident but incorrect search summaries

Especially important if the assistant starts sounding confident while being wrong.

The highest-priority missing AI monitoring

If I were prioritizing, I would implement these first:

Priority 1

Log every AI interaction:

- user query
- expanded query
- model output
- parsed output
- fallback used
- final results shown

Priority 2

Track search outcome:

- clicked result or not
- clarified or not
- abandoned or not
- reformulated query or not

Priority 3

Track extraction quality:

- structured fields extracted



- extraction failures
- correction rate
- null/partial outputs

Priority 4

Track model health:

- latency
 - timeout
 - failure rate
 - queue/load behavior
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Short truth

Marketplace not fully done yet

- teaser/clarification UX polish
- production validation
- broader category intelligence
- trust/seller intelligence
- analytics/observability

AI monitoring not implemented enough

- prompt/output logging
- fallback monitoring
- extraction accuracy tracking
- search quality tracking
- model latency/failure tracking
- human review loop
- drift monitoring



PickPic / SellSell – Engineering Task Board

1. DONE (Completed & Working)

Core AI Search System

- Semantic search using Qdrant is fully integrated
- Hybrid search (Qdrant + PostgreSQL) implemented
- Optimized query flow (limited candidates + $O(1)$ mapping)
- Intent-based dynamic ranking working
- Ontology-based query expansion implemented

Structured Data (Vehicles)

- VehicleDetails schema added to Prisma
- Migration successfully created and applied
- Prisma client generated and working
- Vehicle extraction pipeline integrated into ingestion
- Legacy fallback support maintained

AI Capabilities

- Intent parsing working
- Ambiguity detection implemented
- Guided clarification logic (backend) working
- Session memory using Redis implemented and wired
- Multi-turn query refinement working

Data & Testing

- Real dataset (q84sale) imported successfully
- Mixed data handling (vehicles + non-vehicles) verified
- Search flows tested:
 - Exact match
 - Semantic search
 - Ambiguity flow
 - Session memory
 - Legacy fallback



Performance & Stability

- Node memory optimization completed
- Payload size reduction using Prisma select
- Removal of heavy in-memory sorting loops
- System stable under current load conditions

DevOps / Infrastructure

- Dockerized environment configured
 - PostgreSQL, Redis, Qdrant containers running
 - Migration pipeline fixed (ENUM issue resolved)
 - Entry-point blocking issue fixed (netcat)
 - Container startup flow stabilized
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2. PENDING (Needs Completion / Polish)

Frontend UX (High Visibility)

- Teaser UI (ambiguity results) not properly rendered
 - Currently vertical and clunky
 - Needs horizontal scroll layout
- Quick reply chips (budget, brand suggestions) missing
- Chat flow UX not optimized for mobile

Category Expansion

- Structured schema only exists for vehicles
- No structured extraction for:
 - Phones
 - Electronics
 - Property
 - Furniture
- Ontology limited mainly to vehicle domain

Seller Intelligence Layer

- No listing quality scoring
- No AI-based price recommendation
- No fraud / scam detection (e.g. unrealistic pricing)
- No seller trust or verification scoring



Product & Analytics Visibility

- No search analytics implemented
- No tracking of:
 - clicks
 - conversions
 - abandonment
 - reformulation
- No dashboard for marketplace performance

Local / Dev Environment Finalization

- Docker local browser flow just stabilized, needs final confirmation
- Ensure consistent startup across environments
- Remove deprecated docker-compose version field

3. CRITICAL NEXT (High Priority – Must Be Done Next)

1. Frontend Teaser UX Fix

Implement proper UI for ambiguity results:

- Horizontal scroll container
- Compact product cards
- Maintain chat visibility
- Add quick selection buttons (budget, brand)

This directly impacts user experience.

2. AI Monitoring Layer (Highest Strategic Gap)

Implement logging for every AI interaction:

- User query
- Expanded query (ontology)
- LLM output



- Parsed structured output
- Fallback usage
- Final results returned

Add tracking for:

- extraction success/failure
 - ambiguity trigger rate
 - fallback usage rate
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3. Search Quality Tracking

Track real user behavior:

- query → results count
- query → click
- query → no click
- query → reformulation
- clarification → success rate

This is required to validate AI effectiveness.

4. Model Performance Monitoring

Track system-level AI performance:

- response latency (Ollama)
 - timeout rate
 - failure rate
 - load behavior under concurrency
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5. Extraction Accuracy Monitoring

Track how well AI understands listings:

- % of listings with valid structured data



- missing fields (year, mileage, etc.)
 - incorrect extraction cases
 - fallback vs AI usage ratio
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6. Expand Structured Schema Beyond Vehicles

Next categories to prioritize:

- Phones (brand, model, storage, condition)
- Electronics
- Property

Goal:

Move away from “everything is text” system-wide.

7. Basic Trust & Safety Layer

Implement:

- price anomaly detection (e.g. 1 KWD scams)
 - duplicate listing detection
 - suspicious seller behavior flags
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8. Production Readiness Validation

- Final local browser QA confirmation
 - Deploy to server
 - Run live smoke tests
 - Monitor logs for first real usage
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Final Status Summary

System Maturity: Late Beta (Pre-Production)



What is strong:

- Core AI architecture
- Search intelligence
- Performance and backend stability

What is missing:

- UX polish
- AI observability
- category expansion
- trust layer